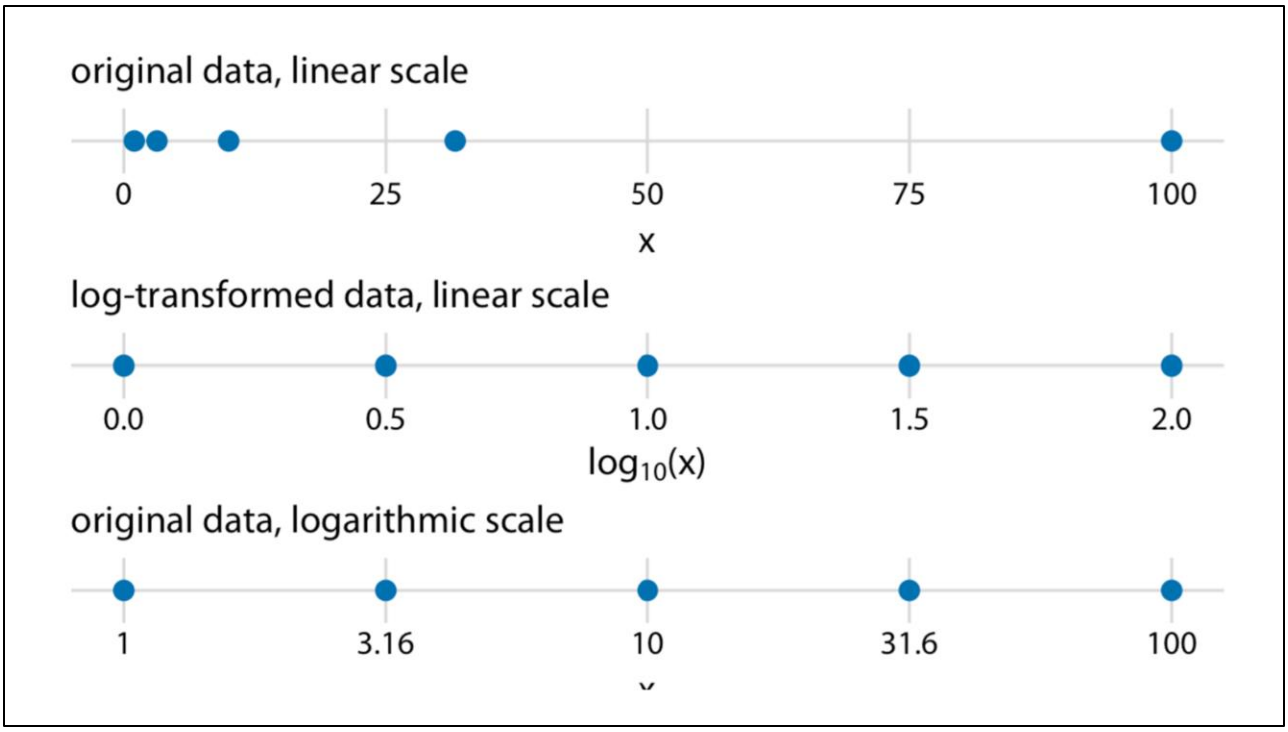
## Linear vs. Nonlinear Scales

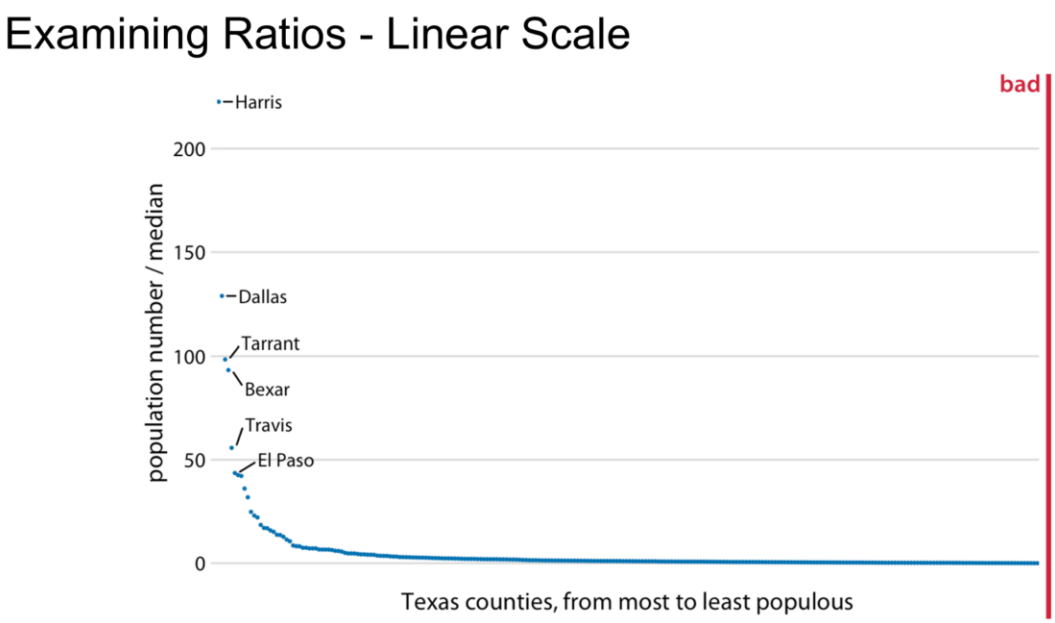
When the grid lines along an axis are spaced evenly both in data units and in the resulting visualization, we are working with a **linear scale**.

In a **nonlinear scale**, even spacing in data units corresponds to uneven spacing in the visualization, or conversely even spacing in the visualization corresponds to uneven spacing in data units.

- Linear scale
    - Data and units are evenly spaced
- Non-linear scales
    - Data or units are unevenly spaced
        - Log scales



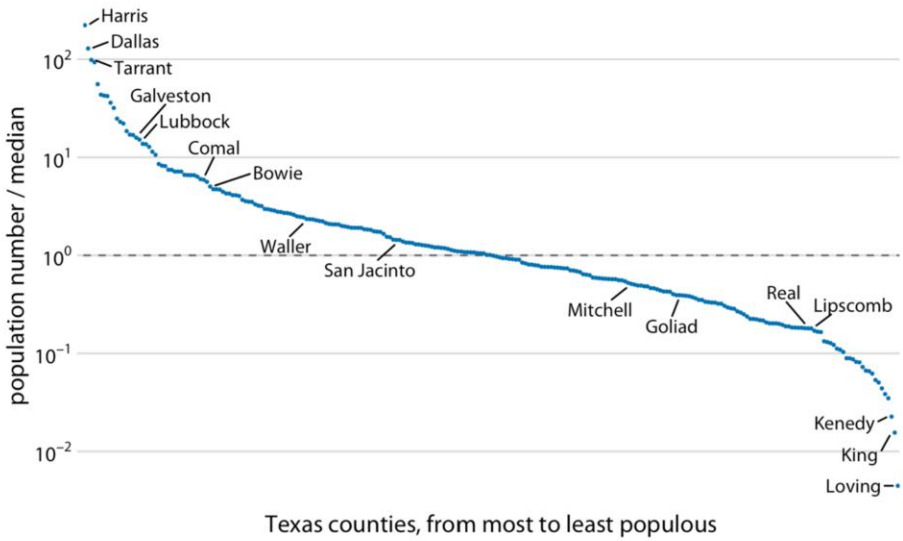

- Linear scale:
  - Data spacing portions match the axis point marker scale
- Log scale data
  - Point position is even but axis data points are not ratioly spaced




- Population number : population per county
- Median : median number of all values of dataset

- Bad
- Values at 0 unable to determine if value is 0 or close to 0

# Examining Ratios - Log Scale




- Y axis scale based on a percentage around the value of 1 at the center
  - Center value represents the median value
  - Small changes are visable